

INTERN INFOBYTE

DEVOPS

TASK-02

Check Your Internship Project x Intern Infobyte Internship Task x Dashboard | EC2 | us-east-1 x CI Pipeline with GitHub Actions x + Ask Gemini x

canva.com/design/DAHA1oKAGP8/pVfzh-VgkkYB3anBV4Nm7w/edit Ask Google ☆ ☆ 🔍 👤 ⋮

Apps ☰ | All Bookmarks

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TASK-2

CI/CD PIPELINE WITH JENKINS

Set up Jenkins to automate build, test, and deployment stages. Integrate GitHub, write a Jenkinsfile, and deploy to a local or cloud server.

Free Resources:

- Jenkins Documentation
- YouTube: Jenkins Full Course – FreeCodeCamp

64 65 CLOUD COMPUTING 66 67 68 69 DEVOPS 70 71 72 73 BLOCKCHAIN TECHNOLOGY

33% Pages 71 / 85 ENG IN 19:37 19-03-2026

Instances | EC2 | us-west-1

CI Pipeline with GitHub Actions

Ask Gemini

us-west-1.console.aws.amazon.com/ec2/home?region=us-west-1#instances:v=3;\$case=tags:true%5C.client:false;\$regex=tags:false%5C.client:false

Apps

shrutu (4132-4059-7866)

shruti

S3EC2VPCIAMAurora and RDSCloudWatchCloudFormationSimple Queue ServiceAmazon RedshiftElastic BeanstalkLambdaRoute 53

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instances Info

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

NameInstance IDInstance stateInstance typeStatus checkAlarm statusAvailability

No instances

You do not have any instances in this region

Launch instances

Select an instance

CloudShell

Feedback

Console Mobile App

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Type here to search

20°C Partly cloudy

ENG IN

19:30

20-03-2026

Launch an instance | EC2 | us-w

CI Pipeline with GitHub Actions

Ask Gemini

us-west-1.console.aws.amazon.com/ec2/home?region=us-west-1#LaunchInstances:

Apps

shruti (4132-4059-7866)

shruti

S3EC2VPCIAMAurora and RDSCloudWatchCloudFormationSimple Queue ServiceAmazon RedshiftElastic BeanstalkLambdaRoute 53

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

task2

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Quick Start

▼ Summary

Number of instances Info

1

Software Image (AMI)

-

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Cancel

Launch instance

Preview code

Launch an instance | EC2 | us-west-1

CI Pipeline with GitHub Actions

Ask Gemini

us-west-1.console.aws.amazon.com/ec2/home?region=us-west-1#LaunchInstances:

Apps

shrutiti (4132-4059-7866)

shrutiti

S3EC2VPCIAMAurora and RDSCloudWatchCloudFormationSimple Queue ServiceAmazon RedshiftElastic BeanstalkLambdaRoute 53

EC2 > Instances > Launch an instance

Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon Linux

ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0507f5acd9ba8e6b7 (64-bit (x86)) / ami-0ca76f453ec67aff1 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0507f5acd9ba8e6b7

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Cancel

Launch instance

Preview code

CloudShellFeedbackConsole Mobile App

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Instances | EC2 | us-west-1

CI Pipeline with GitHub Actions

Ask Gemini

us-west-1.console.aws.amazon.com/ec2/home?region=us-west-1#Instances:v=3;\$case=tags:true%5Cclient:false;\$regex=tags:false%5Cclient:false

Apps

shrutiti (4132-4059-7866)

shrutiti

S3EC2VPCIAMAurora and RDSCloudWatchCloudFormationSimple Queue ServiceAmazon RedshiftElastic BeanstalkLambdaRoute 53

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

1

☒

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability

☒

task2

i-00e5c19da548aa161

Running

t3.micro

Initializing

View alarms +

us-west-1a

i-00e5c19da548aa161 (task2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary Info

Instance ID

i-00e5c19da548aa161

Public IPv4 address

54.219.185.96

Private IPv4 addresses

172.31.28.174

CloudShellFeedbackConsole Mobile App

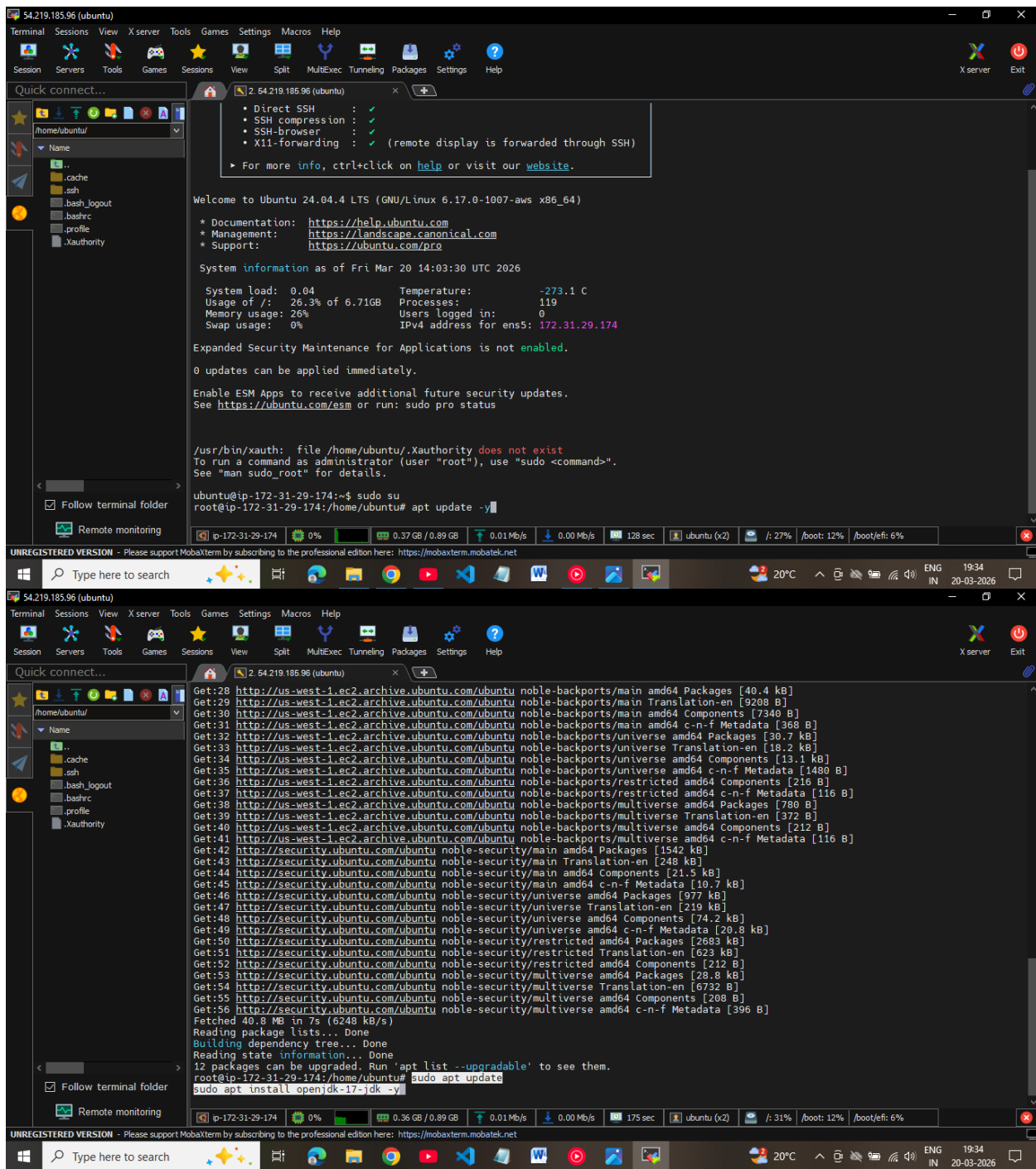
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ENG IN

19:32

20-03-2026



> User Documentation Home

User Handbook

- User Handbook Overview
- Installing Jenkins
 - Docker
 - Kubernetes
 - Linux
 - macOS
 - Windows
 - Other Systems
 - WAR file
 - Other Servlet Containers
 - Offline Installations
 - Initial Settings
- Platform Information
- Using Jenkins
- Pipeline
- Blue Ocean
- Managing Jenkins
- Securing Jenkins
- System Administration

completing the Java installation, continue with the Jenkins installation steps below.

Long Term Support release

A LTS (Long-Term Support) release is chosen every 12 weeks from the stream of regular releases as the stable release for that time period. It can be installed from the [debian-stable apt repository](#).

```
sudo wget -O /etc/apt/keys/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keys/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

BASH |

Weekly release

A new release is produced weekly to deliver bug fixes and features to users and plugin developers. It can be installed from the [debian apt repository](#).

```
sudo wget -O /etc/apt/keys/jenkins-keyring.asc \
https://pkg.jenkins.io/debian/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keys/jenkins-keyring.asc] \
```

54.219.185.96 (ubuntu)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect... 2. 54.219.185.96 (ubuntu)

update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jinfo to provide /usr/bin/jinfo (jinfo) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jlink to provide /usr/bin/jlink (jlink) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmap to provide /usr/bin/jmap (jmap) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmod to provide /usr/bin/jmod (jmod) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jps to provide /usr/bin/jps (jps) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jrunscript to provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jshell to provide /usr/bin/jshell (jshell) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-17-jre:amd64 (17.0.18+8-1~24.04.1) ...
Setting up openjdk-17-jdk:amd64 (17.0.18+8-1~24.04.1) ...
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

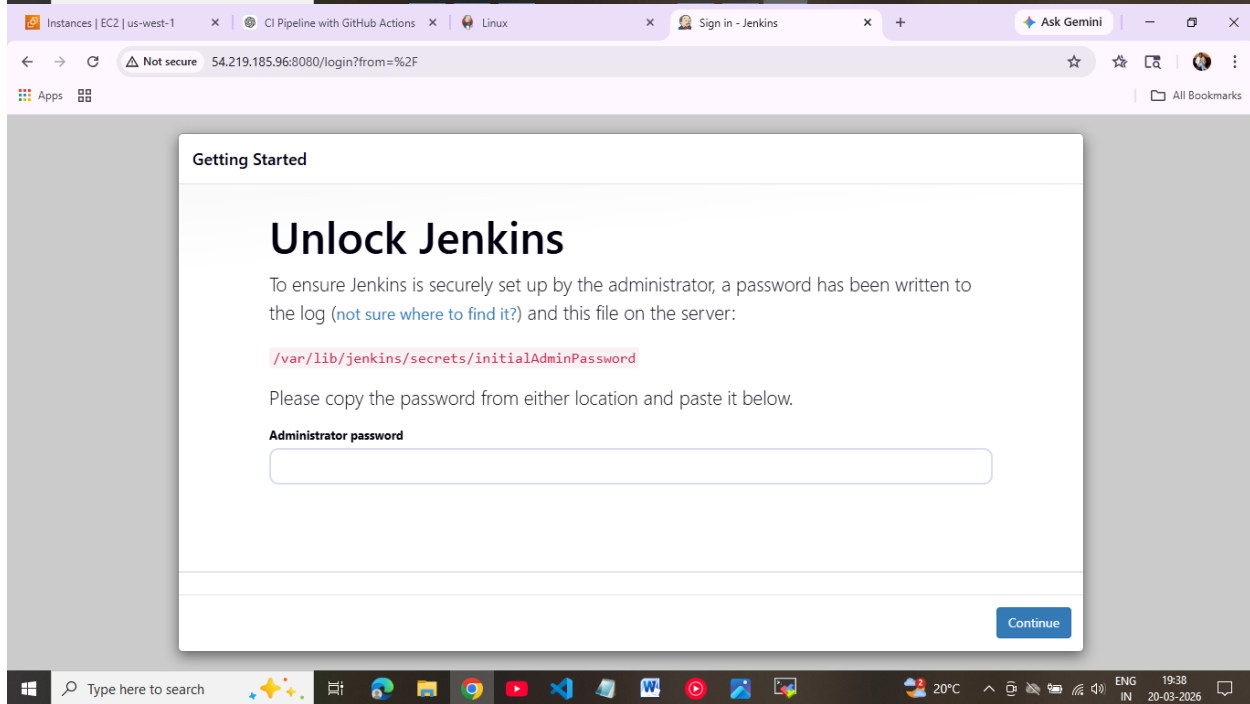
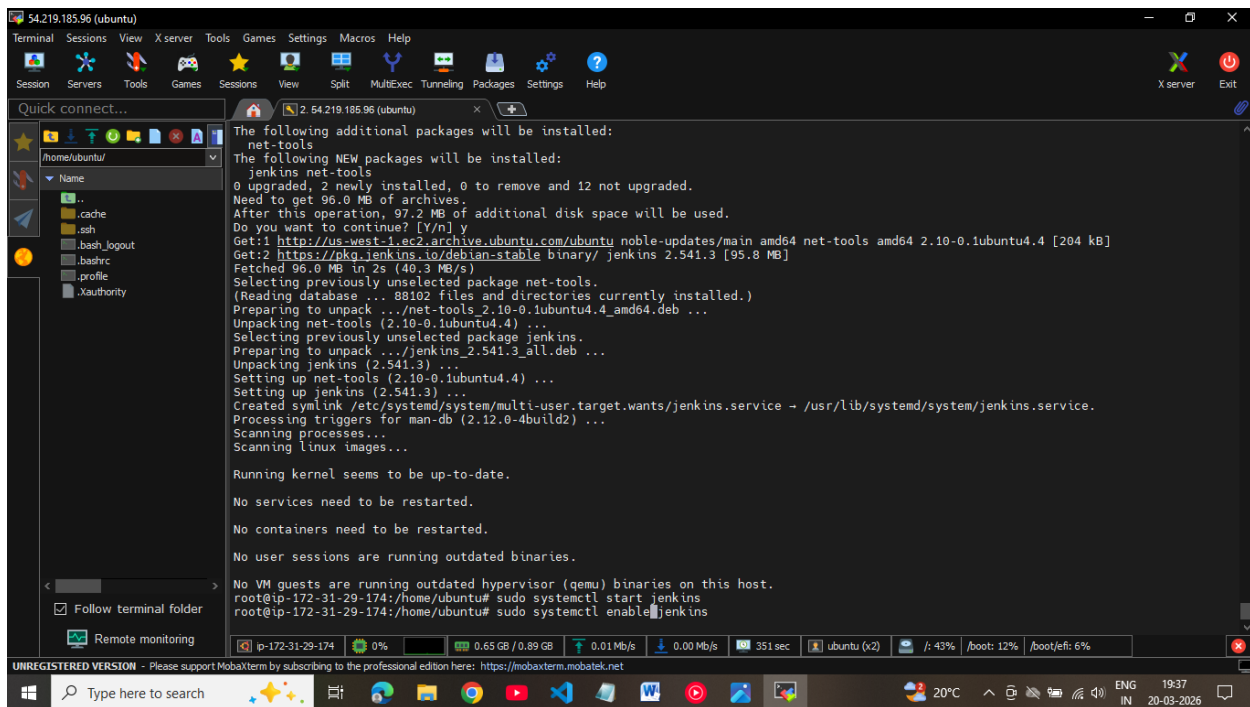
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-29-174:/home/ubuntu# sudo wget -O /etc/apt/keys/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keys/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins

ip-172-31-29-174 0% 0.41 GB / 0.89 GB 0.01 Mb/s 0.01 Mb/s 255 sec ubuntu (x2) /: 40% /boot: 12% /boot/efi: 6%

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54.219.185.96 (ubuntu)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

2 54.219.185.96 (ubuntu)

jenkins net-tools
0 upgraded, 2 newly installed, 0 to remove and 12 not upgraded.
Need to get 96.0 MB of archives.
After this operation, 97.2 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-west-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.541.3 [95.8 MB]
Fetched 96.0 MB in 2s (40.3 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 88102 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.541.3_all.deb ...
Unpacking jenkins (2.541.3) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.541.3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-29-174:/home/ubuntu# sudo systemctl start jenkins
root@ip-172-31-29-174:/home/ubuntu# sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@ip-172-31-29-174:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword

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54.219.185.96:8080

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Select plugins to install

Jenkins 2.541.3

54.219.185.96:8080/#

Instances | EC2 | us-west-1 x CI Pipeline with GitHub Actions x Linux x Setup Wizard - Jenkins x Ask Gemini

Not secure 54.219.185.96:8080

Getting Started

✓ Folders	✓ OWASP Markup Formatter	Build timeout	Credentials binding	Commons Lang Jenkins API
Timestampers	Workspace Cleanup	Ant	Gradle	Folders
Pipeline	GitHub Branch Source	Pipeline: GitHub Groovy Libraries	Pipeline Graph View	OWASP Markup Formatter
Git	SSH Build Agents	Matrix Authorization Strategy	LDAP	** ASX API
Email Extension	Mailer	Dark Theme		** JSON Path API
				** Structs
				** Pipeline: Step API
				** commons-text API
				** Token Macro

** - required dependency

Jenkins 2.541.3

Instances | EC2 | us-west-1 x CI Pipeline with GitHub Actions x Linux x Setup Wizard - Jenkins x Ask Gemini

Not secure 54.219.185.96:8080

Getting Started

Create First Admin User

Username

admin

Password

...

Confirm password

...

Full name

Jenkins 2.541.3

Skip and continue as admin

Save and Continue

The image shows two screenshots of the Jenkins web interface. The top screenshot is the 'Getting Started' page for 'Instance Configuration'. It features a large heading 'Instance Configuration' and a text input field for the 'Jenkins URL' containing 'http://54.219.185.96:8080/'. Below the input field, there is explanatory text about the Jenkins URL and a 'Save and Finish' button. The bottom screenshot is the 'Jenkins Dashboard'. It shows the 'Jenkins' logo, a 'New Item' button, and a 'Build Queue' section indicating 'No builds in the queue.' The main content area displays 'Welcome to Jenkins!' and instructions on how to start building software projects, including buttons for 'Create a job', 'Set up a distributed build', 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'.

Instances | EC2 | us-west-1 x CI Pipeline with GitHub Actions x Linux x New Item - Jenkins x Ask Gemini

Not secure 54.219.185.96:8080/view/all/newJob

Jenkins / All / New Item

New Item

Enter an item name

task2-cicd-pipeline

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**

OK

Instances | EC2 | us-west-1 x CI Pipeline with GitHub Actions x Linux x task2-cicd-pipeline Config - Jenkins x Ask Gemini

Not secure 54.219.185.96:8080/job/task2-cicd-pipeline/configure

Jenkins / task2-cicd-pipeline / Configure

Configure

- General
- Triggers
- Pipeline**
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Pipeline script

Pipeline script from SCM

1

try sample Pipeline...

Save Apply

task2-cicd-pipeline Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/shreyasahay10/intern-infobyte.git

Please enter Git repository.

Save Apply

task2-cicd-pipeline

task2-cicd-pipeline

Permalinks

Add description

Status

Changes

Build Now

Configure

Delete Pipeline

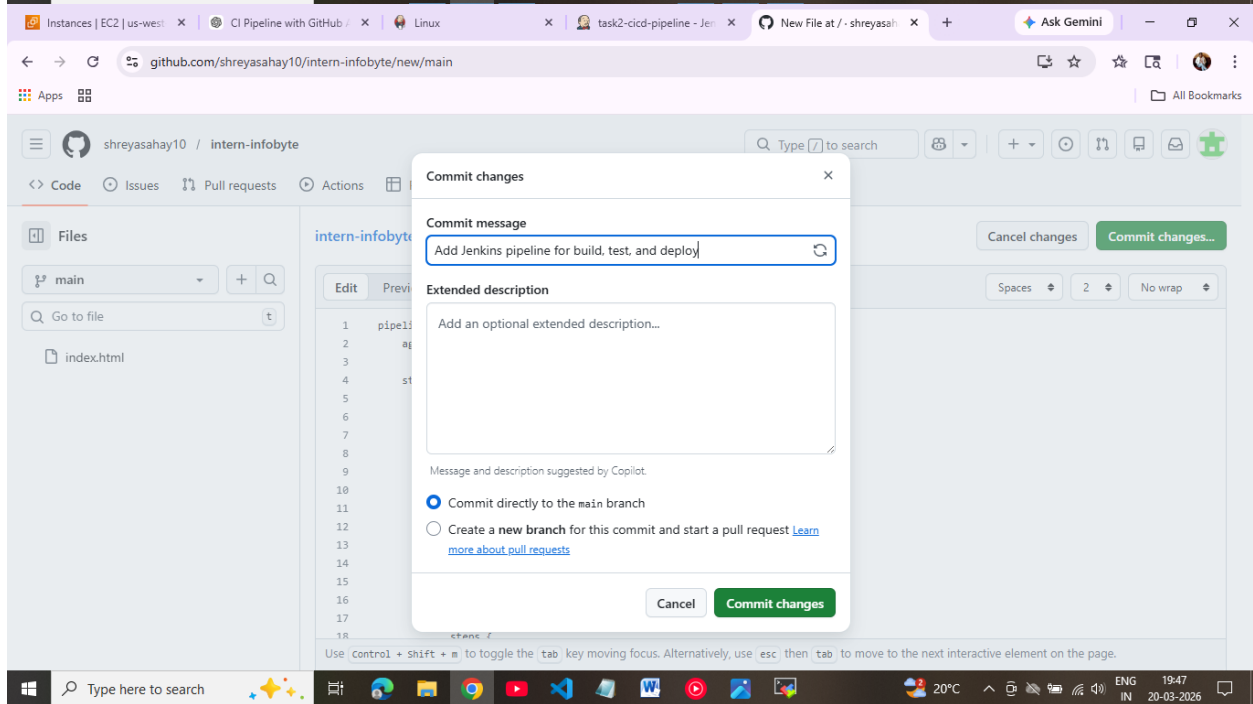
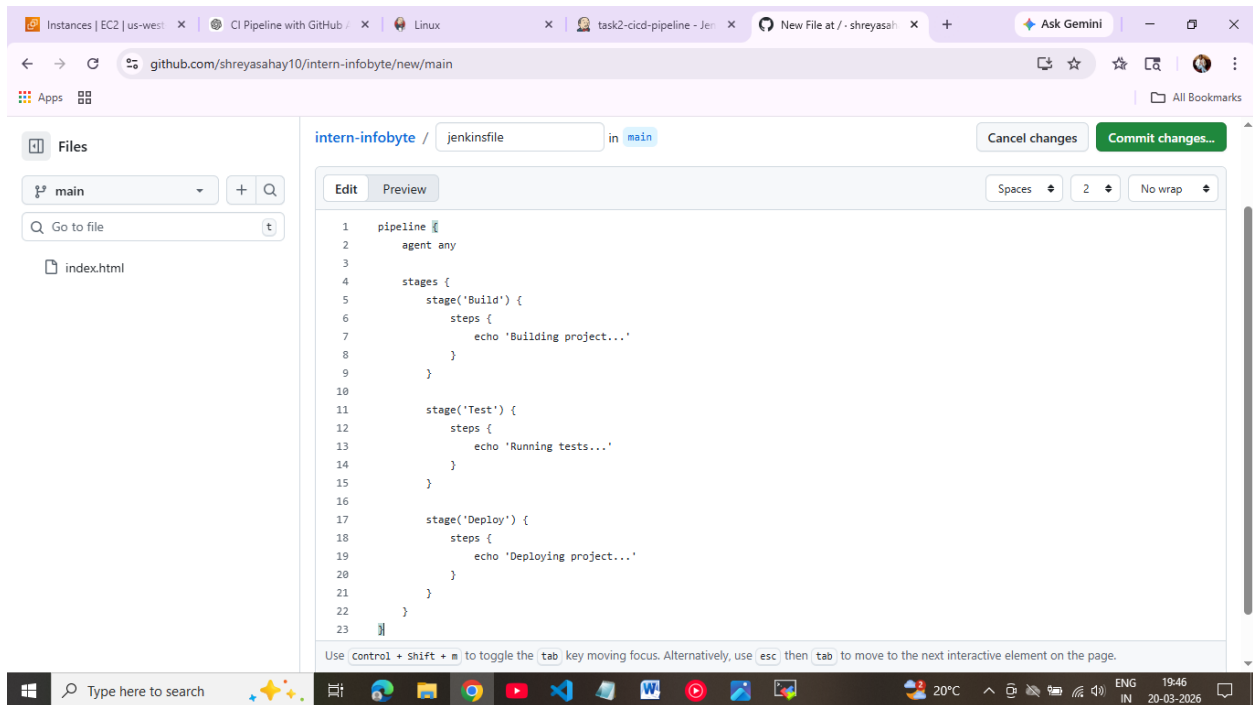
Stages

Rename

Pipeline Syntax

Builds >

No builds



Jenkins / task2-cicd-pipeline / #2

Status ✓ #2 (20-Mar-2026, 2:20:16 pm) Add description Keep this build forever

Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Started by user admin

Started 12 sec ago
Took 6.7 sec

This run spent:

- 20 ms waiting;
- 6.7 sec build duration;
- 6.7 sec total from scheduled to completion.

git Revision: 160ce76697e230709563c70efac2c79bf2135404
Repository: <https://github.com/shreyasahay10/intern-infobyte.git>

- refs/remotes/origin/main

No changes.

Jenkins / task2-cicd-pipeline / #2

Console Download Copy View as plain text

```
Started by user admin
Obtained Jenkinsfile from git https://github.com/shreyasahay10/intern-infobyte.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/task2-cicd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/shreyasahay10/intern-infobyte.git
> git init /var/lib/jenkins/workspace/task2-cicd-pipeline # timeout=10
Fetching upstream changes from https://github.com/shreyasahay10/intern-infobyte.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/shreyasahay10/intern-infobyte.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
```

Instances | EC2 | us-west | x CI Pipeline with GitHub / x Linux x Console - #2 - Jenkins x intern-infobyte/Jenkinsf x + Ask Gemini - x

← → ↻ Not secure 54.219.185.96:8080/job/task2-cicd-pipeline/2/console ☆ ☆ [🔍] [👤] : All Bookmarks

Apps [🔍] [⚙️] [👤]

Jenkins / task2-cicd-pipeline / #2

```
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 160ce76697e230709563c70efac2c79bf2135404 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 160ce76697e230709563c70efac2c79bf2135404 # timeout=10
Commit message: "Rename jenkinsfile to Jenkinsfile"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] }
[Pipeline] // stage
```

Windows Taskbar: Type here to search, 20°C, ENG IN, 19:52, 20-03-2026

Instances | EC2 | us-west | x CI Pipeline with GitHub / x Linux x Console - #2 - Jenkins x intern-infobyte/Jenkinsf x + Ask Gemini - x

← → ↻ Not secure 54.219.185.96:8080/job/task2-cicd-pipeline/2/console ☆ ☆ [🔍] [👤] : All Bookmarks

Apps [🔍] [⚙️] [👤]

Jenkins / task2-cicd-pipeline / #2

```
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.3

Windows Taskbar: Type here to search, 20°C, ENG IN, 19:52, 20-03-2026